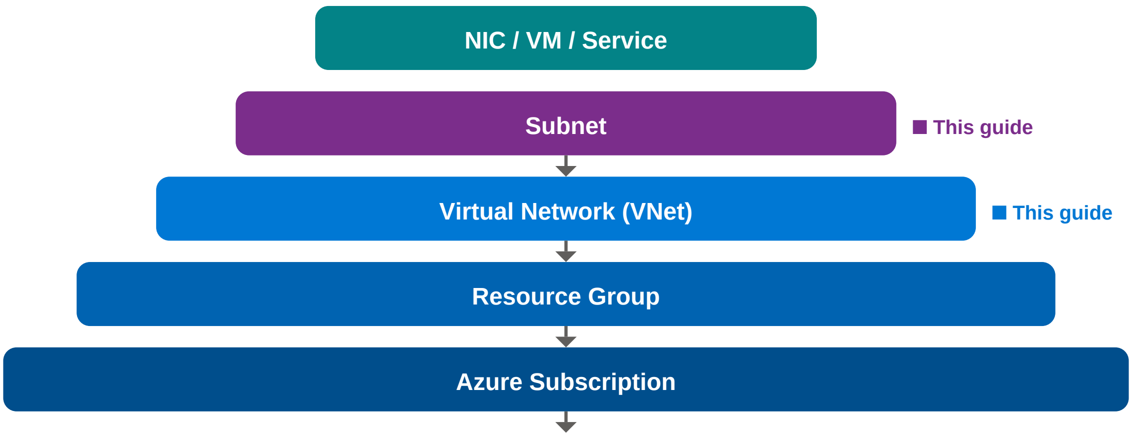
VNet vs Subnet

Azure Cloud Networking — Complete Visual Guide

Definitions · Architecture Diagrams · Head-to-Head Comparison · Real-World Scenarios

1 Azure Networking Hierarchy

Before comparing VNet and Subnet, it helps to see where each sits in the Azure resource hierarchy. A **VNet** is the broadest networking boundary you create in Azure, while a **Subnet** is a subdivision within that VNet.



Every Azure VM, container, or managed service must be placed inside a Subnet — and every Subnet must belong to a VNet.

2 Definitions

Virtual Network (VNet)

A **VNet** is the fundamental private network boundary in Microsoft Azure. It is a logically isolated network that you fully control — including IP address space, DNS settings, security policies, and routing. A VNet is scoped to a single **Azure Region** but can span multiple **Availability Zones** within that region. VNets can be connected to each other (VNet Peering), to on-premises networks (VPN Gateway / ExpressRoute), and to the public internet.

Example CIDR: `10.0.0.0 / 16` -> 65,536 total IPs available for all resources in this VNet

Region-scoped	Peerable	Custom DNS	Private IP Space

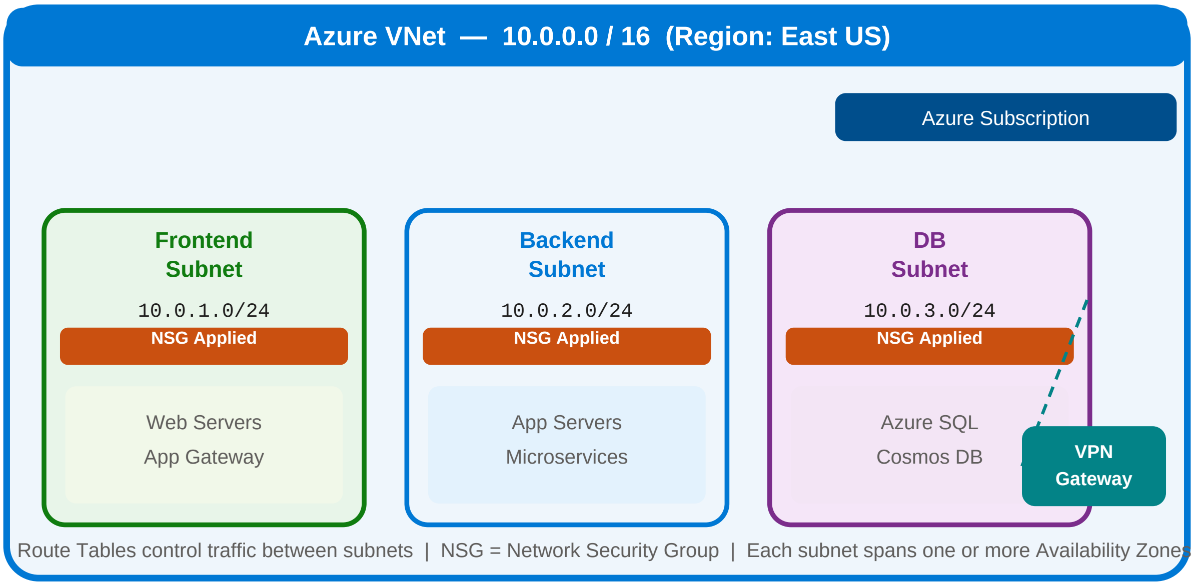
Subnet

A **Subnet** is a range of IP addresses carved out of a VNet's address space. Each Azure resource (VM, App Service Environment, AKS node pool, etc.) is deployed into a specific subnet. Subnets enable you to **segment** your VNet for security, performance, and operational isolation. You can attach a **Network Security Group (NSG)** and a custom **Route Table** to each subnet independently.

Example CIDR: `10.0.1.0 / 24` -> 251 usable IPs (Azure reserves 5 per subnet)

NSG per Subnet	Route Table	Service Endpoints	Delegation

3 Architecture Diagram



Above: A single Azure VNet (10.0.0.0/16) with three subnets segregating the **frontend**, **backend**, and **database** tiers. Each subnet has its own NSG controlling inbound/outbound traffic.

4 Head-to-Head Visual

VNet	VS	Subnet
Entire cloud network boundary Spans a full Azure Region CIDR: /8 to /29 (up to /16 typical) Peered with other VNets / on-prem		Segment inside a VNet Tied to specific Availability Zone(s) CIDR must be subset of VNet CIDR Has own NSG + Route Table

Feature	VNet	Subnet
Scope	Entire Azure Region	Part of a VNet, 1+ Availability Zones
CIDR Size	Up to /8 (millions of IPs)	Must be subset; /24-/28 typical
Purpose	Network isolation boundary	Resource segmentation & access control
Security Layer	Azure Firewall / DDoS Protection	Network Security Group (NSG)
Routing	Between VNets or to on-prem	Route Table per subnet
Connectivity	VNet Peering, VPN, ExpressRoute	Service Endpoints, Private Endpoints
Azure Reserves	Nothing - full range yours	5 IPs per subnet (first 4 + last 1)
Delegation	Not applicable	Delegate to Azure services (e.g. ACI)
DNS	Custom DNS at VNet level	Inherits VNet DNS (can override)
Billing	Free (pay for gateways/peers)	Free (pay for NAT, Private Endpoints)

5 Real-World Azure Scenarios

■ Banking Application (3-Tier Architecture)

VNet:	`10.10.0.0/16 — Production VNet (East US)`
■ **GatewaySubnet**	`10.10.1.0/24` — VPN / ExpressRoute Gateway — connects to on-premises data center
■ **frontend-subnet**	`10.10.2.0/24` — Azure App Gateway + Web VMs — public-facing, NSG allows port 443 only
■ **backend-subnet**	`10.10.3.0/24` — Java App Servers — no public IPs, talks only to frontend + DB subnets
■ **db-subnet**	`10.10.4.0/24` — Azure SQL Managed Instance — Private Endpoint, no internet access
Key Insight:	Regulatory compliance (PCI-DSS) requires strict isolation. The DB subnet has zero public routes — only the backend subnet can reach it via NSG rules.

■ E-Commerce Platform on AKS (Kubernetes)

VNet:	`172.16.0.0/16 — AKS Production VNet`
■ **aks-system-subnet**	`172.16.1.0/22` — AKS System Node Pool — 1,022 IPs for cluster overhead
■ **aks-user-subnet**	`172.16.5.0/22` — AKS User Node Pool — 1,022 IPs for workload pods
■ **ingress-subnet**	`172.16.9.0/24` — Nginx Ingress Controller + Azure Load Balancer
■ **monitoring-subnet**	`172.16.10.0/24` — Prometheus, Grafana, Log Analytics Workspace
Key Insight:	AKS with Azure CNI needs large subnets (/22) because each Pod gets its own VNet IP. A /24 would only support ~50 pods — plan for growth!

■ Multi-Region Hub-and-Spoke Network	
VNet:	`10.0.0.0/16 — Hub VNet (peered with 4 Spoke VNets)`
■ **AzureFirewallSubnet**	`10.0.0.0/26` — REQUIRED name for Azure Firewall — inspects all spoke traffic
■ **AzureBastionSubnet**	`10.0.0.64/27` — REQUIRED name for Azure Bastion — secure RDP/SSH, no public IPs
■ **GatewaySubnet**	`10.0.0.96/27` — VPN + ExpressRoute Gateway — on-prem connectivity
■ **shared-services**	`10.0.1.0/24` — DNS, AD Domain Controllers, SIEM tools
Key Insight:	Azure enforces exact subnet names for Firewall (AzureFirewallSubnet) and Bastion (AzureBastionSubnet). These are not flexible — a naming mistake means the service won't deploy.

6 Azure-Specific Rules & Gotchas

■■ **Azure reserves 5 IPs per subnet**

First 4 addresses + last address in every subnet are reserved by Azure (network, gateway, DNS x2, broadcast). A /28 gives only 11 usable IPs, not 16.

■ **NSG can be attached at Subnet OR NIC level**

You can stack NSGs — one at the subnet level (broad rules) and one per NIC (fine-grained). If both exist, traffic must pass BOTH NSGs inbound and outbound.

■ **Some subnet names are reserved by Azure**

AzureFirewallSubnet, AzureBastionSubnet, GatewaySubnet — these exact names are required for those services. You cannot rename them after creation.

■ **VNets cannot span regions — Subnets cannot span VNets**

A VNet is regional. To connect two regions, use VNet Peering or a VPN. A Subnet must stay within its parent VNet's address space.

■ **Address space can be expanded but not easily shrunk**

You can add additional CIDR blocks to a VNet later, but removing or modifying existing ranges requires deleting deployed resources first. Plan ahead.

7 Key Takeaways

- VNet = your private datacenter in Azure. Everything networking starts here.
- Subnet = a zone within that datacenter. Each tier of your app gets its own subnet.
- Security lives at the Subnet level via NSGs. Think of subnets as security perimeters.
- Always over-provision your VNet CIDR. You can't easily shrink it later.
- For AKS / PaaS delegation, check IP requirements first — pods eat IPs fast.
- Azure reserves 5 IPs per subnet automatically — subtract those from your capacity math.